

An exact fixed-point formula for the consistent Bellman operator

Automated Math Discovery Run fa_banach_001
Model: GPT5.6

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Claimed status: full quantitative answer, subject to human review. The formula below completely determines the consistent Bellman fixed point from the classical optimal Q -function and the one-step self-transition probabilities.

1 Source question

Kadurha David asks how the unique fixed point of the consistent Bellman operator relates to the fixed point obtained from the classical Bellman operator [1, Chapter 5, Section 5.1.2, printed p. 34]. The passage appears on PDF page 38:

Section 5.1. Alternatives to the Bellman Operator

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From this proof, we can conclude that $\mathcal{T}_c$ has only one fixed point and that fixed point is optimal for this setup (that is the setup in which we are using the consistent Bellman equation instead of the classical one).

The question that remains is how this fixed point relates to the one we have found using the classical Bellman operator.

The most important thing is that, we are sure to converge to a unique fixed-point. As we don't yet have the mathematical tool to say something about the relation between that fixed point and the one which will be found using the classical Bellman Operator, we can just wait and see how the operator will behave in practice in comparison to the Bellman Operator.

Figure 1: The open question in the source paper.

Bellemare et al. had already proved the qualitative facts that the consistent operator is optimality-preserving and gap-increasing [2, Corollary 3]. The result here gives the missing exact coordinatewise relation.

2 Setting and notation

Let $\mathcal{S}$ and $\mathcal{A}$ be nonempty finite sets, let $P(\cdot \mid s, a)$ be transition probabilities, let $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, and let $0 \leq \gamma < 1$. For a Q -function q , put $v_q(s) = \max_{a \in \mathcal{A}} q(s, a)$. The classical and consistent

Bellman operators are

$$(\mathcal{T}q)(s, a) = r(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x \mid s, a) v_q(x),$$

$$(\mathcal{T}_c q)(s, a) = r(s, a) + \gamma \left(P(s \mid s, a) q(s, a) + \sum_{x \neq s} P(x \mid s, a) v_q(x) \right).$$

Both are γ -contractions in the supremum norm. Write q^* for the fixed point of $\mathcal{T}$, q_c for the fixed point of $\mathcal{T}_c$, and

$$v^*(s) = \max_a q^*(s, a), \quad d(s, a) = P(s \mid s, a).$$

3 Proof intuition

At a consistent fixed point, the term caused by returning immediately to the same state contains $q_c(s, a)$ itself. Moving that term to the left eliminates the self-loop and produces an operator on state values alone. Its contraction factor is no larger than γ , even after division by $1 - \gamma d(s, a)$. The classical optimal value v^* is a fixed point of this new operator: every action gives at most $v^*(s)$, and a classical optimal action gives equality. Uniqueness therefore forces $\max_a q_c(s, a) = v^*(s)$. Substitution then yields the exact formula, not just an inequality between action gaps.

4 Exact relation

Theorem 1 (Coordinatewise fixed-point formula). *For every $(s, a) \in \mathcal{S} \times \mathcal{A}$,*

$$\boxed{v^*(s) - q_c(s, a) = \frac{v^*(s) - q^*(s, a)}{1 - \gamma d(s, a)}}. \quad (1)$$

Equivalently,

$$q_c(s, a) = \frac{q^*(s, a) - \gamma d(s, a) v^*(s)}{1 - \gamma d(s, a)}. \quad (2)$$

In particular, $\max_a q_c(s, a) = v^(s)$ for every state.*

Proof. For a bounded function $v : \mathcal{S} \rightarrow \mathbb{R}$, define

$$H_a v(s) = \frac{r(s, a) + \gamma \sum_{x \neq s} P(x \mid s, a) v(x)}{1 - \gamma d(s, a)},$$

$$Hv(s) = \max_{a \in \mathcal{A}} H_a v(s).$$

The denominator is positive. For each s, a ,

$$|H_a v(s) - H_a w(s)| \leq \frac{\gamma(1 - d(s, a))}{1 - \gamma d(s, a)} \|v - w\|_\infty$$

$$\leq \gamma \|v - w\|_\infty,$$

because $1 - d \leq 1 - \gamma d$. Taking maxima does not increase this bound, so H is a γ -contraction and has a unique fixed point.

Put $v_c(s) = \max_a q_c(s, a)$. The equation $\mathcal{T}_c q_c = q_c$, with its self-loop term moved to the left, says exactly

$$q_c(s, a) = H_a v_c(s).$$

Taking the maximum over a gives $v_c = H v_c$.

We next show that v^* is also an H -fixed point. The classical fixed-point equation gives

$$q^*(s, a) = r(s, a) + \gamma d(s, a) v^*(s) + \gamma \sum_{x \neq s} P(x \mid s, a) v^*(x).$$

Consequently,

$$v^*(s) - H_a v^*(s) = \frac{v^*(s) - q^*(s, a)}{1 - \gamma d(s, a)} \geq 0. \quad (3)$$

Since $\mathcal{A}$ is finite, some action a^* attains $q^*(s, a^*) = v^*(s)$; for this action equality holds in (3). Hence $H v^* = v^*$. Uniqueness of the H -fixed point now yields $v_c = v^*$.

Finally, $q_c(s, a) = H_a v^*(s)$. Replacing the numerator of $H_a v^*$ by $q^*(s, a) - \gamma d(s, a) v^*(s)$ proves (2); subtracting it from $v^*(s)$ gives (1). $\square$

Corollary 2 (Exact policy and gap consequences). *For every state s :*

1. $\arg \max_a q_c(s, a) = \arg \max_a q^*(s, a)$, including all ties;
2. each action gap is amplified by the factor $(1 - \gamma d(s, a))^{-1} \in [1, (1 - \gamma)^{-1}]$;
3. $q_c(s, a) \leq q^*(s, a)$, and the inequality is strict exactly when $q^*(s, a) < v^*(s)$ and $d(s, a) > 0$.

Proof. The first two assertions follow immediately from (1). For the third, subtract (2) from $q^*(s, a)$ to obtain

$$q^*(s, a) - q_c(s, a) = \frac{\gamma d(s, a)}{1 - \gamma d(s, a)} (v^*(s) - q^*(s, a)).$$

$\square$

5 Verification and review notes

The script `code/check_random_mdps.py` generated 400 random finite MDPs (two to seven states, two to five actions), mixed in self-loop weights up to 0.995, and independently iterated both operators. The worst observed direct formula error was 1.08×10^{-11} ; the worst residual after substituting the formula into $\mathcal{T}_c$ was 2.07×10^{-13} ; and the maximum state values agreed to displayed floating-point precision. These checks are not part of the proof.

The theorem is deliberately stated for finite tabular MDPs, matching the source’s summation formulas and attained maxima. The proof extends to broader bounded settings under the usual measurability and supremum hypotheses, but that extension is not claimed here. The main human-review point is the reduction from the Q -fixed-point equation to the contraction H .

6 Bounded novelty check

The run’s registry, solution, attempt, and proof-gap indexes had no hit for arXiv:2410.03706 or this fixed-point relation. Searches over arXiv, AAAI, NeurIPS, and OpenReview combined “consistent Bellman operator” with “fixed point”, “formula”, “self transition”, and “action gap”. The primary qualitative predecessor [2] and later reviews were inspected. They establish optimality preservation

and gap increase, but the explicit coordinate identity (1) was not found. This was a bounded, not exhaustive, search; novelty confidence is therefore moderate and is limited to the exact formula and its sharp consequences.

References

- [1] K. Kadurha David, *Topological Foundations of Reinforcement Learning*, arXiv:2410.03706, 2024. <https://arxiv.org/abs/2410.03706>
- [2] M. G. Bellemare, G. Ostrovski, A. Guez, P. S. Thomas, and R. Munos, *Increasing the Action Gap: New Operators for Reinforcement Learning*, Proc. AAAI Conf. Artificial Intelligence 30 (2016), 1476–1483; arXiv:1512.04860. <https://doi.org/10.1609/aaai.v30i1.10303>